

Introduction to Chemistry

AI Tutor - Chemistry - Lesson 1

Introduction

Chemistry is the study of matter: what it's made of, how it behaves, and how it changes when it interacts with other matter. All matter exists in one of three common states (solid,...

Explanation

Chemistry is the study of matter: what it's made of, how it behaves, and how it changes when it interacts with other matter. All matter exists in one of three common states (solid, liquid, gas), and changes can be physical (like ice melting, where the substance stays the same) or chemical (like wood burning, which creates entirely new substances). Chemistry bridges physics and biology, explaining everything from cooking to medicine to materials science.

Key Points

- Solids have a fixed shape and volume.
- Liquids have a fixed volume but take their container's shape.
- Gases have no fixed shape or volume.
- A physical change alters appearance; a chemical change creates new substances.

Examples

- Ice melting into water is a physical change.
- Wood burning into ash and smoke is a chemical change.
- Water boiling into steam is a physical change (still H₂O).

Summary

Chemistry studies matter's composition and behavior, distinguishing physical changes (appearance only) from chemical changes (new substances formed).

Practice Questions

- 1 Name the three common states of matter.
- 2 Is rusting iron a physical or chemical change? Why?
- 3 Give one example each of a physical and a chemical change.

Quiz

Q1. Which state of matter has no fixed shape or volume?

a) Solid | b) Liquid | c) Gas | d) Plasma

Q2. Ice melting into water is an example of a:

a) Chemical change | b) Physical change | c) Nuclear change | d) Biological change

Q3. Which of these is a chemical change?

a) Water boiling | b) Wood burning | c) Ice melting | d) Sugar dissolving